



Alberta Public Health Disease Management Guidelines

Tetanus



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Communicable Disease Control Branch

Public Health Division

Alberta Health

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Revisions

Revision Date	Document Section	Description of Revision
August 2024	General	Entire guideline reviewed and updated to align with current information.
	Public Health Management	- Updated Key Investigation section to include assessment of immunization history. - Updated Management of a Case section: - Removed details on wound management. - Added a link to the Tetanus Post-exposure Prophylaxis in Injury/Wound Management guidance document.

Case Definition

Confirmed Case

Clinical illness^(A) without other apparent medical cause:

- with or without laboratory evidence of *Clostridium (C.) tetani*

AND

- with or without history of injury.

^(A) Clinical illness is characterized by acute onset of hypertonia and/or painful muscular contractions (usually of the muscles of the jaw and neck), and generalized muscle spasms without other apparent medical cause. See [Clinical Presentation](#) for more information.

Reporting Requirements

Physicians, Health Practitioners and Others

- Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed cases in the prescribed form by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

- All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:
 - MOH (or designate) of the zone, and
 - Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services and First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- For out-of-province and out-of-country cases, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days) including:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Clinical Assessment and Epidemiology

Etiology

Tetanus is caused by a neurotoxin that is produced by anaerobic, spore forming, gram-positive bacilli, *Clostridium (C.) tetani*.

Clinical Presentation

Tetanus is an acute neurological disease caused by an exotoxin of *C. tetani*, which grows anaerobically at the site of an exposure, typically during cutaneous injury. However, an exposure site may not always be apparent.^(1,2)

Tetanus infection can be classified into one of the following:

Generalized tetanus is the most common manifestation and is characterized by painful, severe muscle spasms, often triggered by sensory stimuli. Typical features include trismus or lockjaw, opisthotonos (backward arching of the head, neck, and spine) and risus sardonicus (fixed smile and elevated eyebrows).^(3,4)

Localized tetanus is not as common and manifests as muscles spasms at the site of injury.⁽³⁾ Spasms may be mild and persistent, lasting for weeks, often resolving spontaneously.

Cephalic tetanus is rare and affects the cranial nerves, most commonly the facial nerve, and may be associated with chronic otitis media and/or infected wounds from injuries to the neck or head.⁽³⁻⁵⁾

Neonatal tetanus occurs through contamination of the umbilical cord in infants born to those who are inadequately immunized or unimmunized and when unsterile instruments are used during the delivery process. This type of tetanus is most commonly seen in resource-limited countries.^(6,7)

Complications of any form of tetanus may include the following:

- respiratory failure due to spasm and rigidity of the chest wall and dysfunction of the diaphragm,
- aspiration pneumonia,
- laryngeal spasms that cause airway obstruction and respiratory apnea, and
- broken bones due to severe muscle spasms.⁽³⁾

The most common cause of death is respiratory failure. Most fatalities occur in infants and the elderly, with the highest case fatality rate (ranging from 10–80%) among those who are unimmunized.^(8,9)

Diagnosis

The diagnosis of tetanus is generally based on symptoms and clinical assessment. There are no confirmatory laboratory tests, however, diagnosis can be supported by assessment of immunization status, culture of *C. tetani* from the wound, detection of IgG against tetanus toxoid and detection of circulating tetanus toxin.^(2,3)

C. tetani can only be cultured from infected wounds in approximately 30% of cases. A negative culture result does not rule out infection. Isolation of *C. tetani* from individuals without symptoms is also possible.^(4,10)

Treatment

Cases require hospitalization and should receive tetanus immune globulin (TIG), appropriate wound management, supportive care for tetanic spasms, careful monitoring, airway management and, frequently, mechanical ventilation.^(1,11)

Reservoir

The spores are pervasive in the environment and are mostly found in the soil, the intestines and feces of humans and animals.^(1,3,8)

Transmission

There is no person-to-person transmission. Individuals who are unimmunized or who are inadequately immunized become infected when spores enter the body through a wound (e.g., broken skin caused by cuts, burns, injuries or a puncture/surgical wound), germinate and produce toxin.^(3,11)

Incubation Period

The incubation period is typically 3–21 days but it may range from one day to several months. A shorter incubation period is associated with more severe disease, and poor prognosis.^(1,8)

Period of Communicability

Unlike other vaccine-preventable diseases, tetanus is not communicable.⁽¹²⁾

Host Susceptibility

Susceptibility is universal. Previous tetanus infection does not confer immunity from infection. A primary series followed by booster doses of tetanus toxoid-containing vaccines induce immunity that lasts for at least 10 years after complete immunization, with declining immunity over time.^(9,12)

Incidence

Cases of tetanus are reported worldwide and occur mainly in agricultural regions and areas where tetanus immunization programs are inadequate, and in low-income countries where antenatal care services may be limited.^(1,4)

Tetanus is rare in developed countries such as Canada due to robust immunization programs. The [Notifiable Diseases Online](#) website provides counts and rates of tetanus cases reported in Canada. Tetanus case counts in Alberta are available on the [Interactive Health Data Application \(IHDA\)](#)

Public Health Management

Key Investigation

- Confirm that the individual meets case definition.
- Obtain history of illness including date of onset of signs and symptoms and determine incubation period.
- Determine tetanus specific immunization history:
 - number of doses,
 - date administered,
 - where the person was immunized (e.g., out of country), and
 - if not immunized, determine reason why.
- Determine the possible source of infection, including recent travel history
 - identify any recent injury (e.g., cuts, burns, puncture/surgical wound or laceration).

Management of a Case

- Provide information about disease transmission, prevention, and control.
- Hospitalized cases should be managed in accordance with hospital infection prevention and control guidelines.
- Supportive care and medications to control tetanic spasms.
- Refer to the [Tetanus Post-exposure Prophylaxis in Injury/Wound Management](#) guidance document for information on wound management.
- If eligible, according to the current [Alberta Immunization Policy \(AIP\)](#), cases who are unimmunized or partially immunized should be offered tetanus containing vaccine as infection does not confer immunity from subsequent exposures.

Management of Contacts

- No follow-up is required, as tetanus is not transmitted from person-to-person.

Preventative Measures

- Educate the public on the dangers of tetanus infection and the importance of immunization against this vaccine preventable disease. Emphasize the need for timely immunization, including any necessary travel immunization prior to departure.
- Refer to the [AIP](#) for current immunization recommendations.
- Educate the public on proper wound care: use first aid for minor wounds such as breaks in the skin, scrapes or blisters that are not infected. However, seek medical attention for serious wounds, including burns, frostbite or any wound that is infected or contaminated with soil, feces or dirt.

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